ELEC 231 — Mathematical Methods I for ECE · Fall 2026

Week 4: Exact first order ODEs; introduction to second order ODEs

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This note covers selected content from Chapter 2 of Murray R. Spiegel's *Applied Differential Equations*.

Notation.

j	imaginary unit, $\mathrm{j}^2=-1$	$\mathcal{L}\{f(t)\}$	Laplace transform
$\mathrm{e}^{-t/\tau}$	natural exponential	$\mathcal{L}^{-1}\{F(s)\}$	inverse transform
$\frac{\mathrm{d}y}{\mathrm{d}t}$, $\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$	derivatives	$v*h$	convolution
$u(t)$	unit step	$\boldsymbol{x}$, $\mathbf{A}$	vector, matrix
$\delta(t)$	unit impulse	τ	time constant

1 Exact differential equations (Spiegel, Applied Differential Equations)

First-order ODE

Theorem 4.1 (Exactness). A *necessary* + *sufficient* condition for the *exactness* of the differential equation

$$M\ \mathrm{d}t + N\ \mathrm{d}y = 0 \tag{4.1}$$

is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}. \tag{4.2}$$

Source: Spiegel, *Applied Differential Equations*.

Example 4.1 (The exact equation $(2ty+3t^2)\,\mathrm{d}t + t^2\,\mathrm{d}y = 0$).

$$(2ty+3t^2)\ \mathrm{d}t + t^2\ \mathrm{d}y = 0 \tag{4.3}$$

$$M = 2ty+3t^2, \qquad N = t^2 \tag{4.4}$$

$$\frac{\partial M}{\partial y} = 2t, \qquad \frac{\partial N}{\partial t} = 2t \tag{4.5}$$

By the Theorem, there is a function $U(t,y)$ such that

$$(2ty+3t^2)\ \mathrm{d}t + t^2\ \mathrm{d}y = \mathrm{d}U.$$

From calculus, the differential

$$\mathrm{d}U(t,y) = \frac{\partial U}{\partial t}\ \mathrm{d}t + \frac{\partial U}{\partial y}\ \mathrm{d}y,$$

so

$$\frac{\partial U}{\partial t} = 2ty + 3t^2 \tag{4.6}$$

$$\frac{\partial U}{\partial y} = t^2. \tag{4.7}$$

We seek $U(t, y)$. From (4.6), reverse differentiate (i.e. integrate) with respect to t yields

$$U = t^2 y + t^3 + f(y). \tag{4.8}$$

Here $f(y)$ is a constant of integration, and must be independent of t, but may depend on y. Substituting U from (4.8) into (4.7) yields

$$\frac{\partial}{\partial y}(t^2 y + t^3 + f(y)) = t^2 + \frac{\mathrm{d}f(y)}{\mathrm{d}y} = t^2 \qquad \textit{only the } f(y) \textit{ term survives besides } t^2$$

$$\frac{\mathrm{d}f(y)}{\mathrm{d}y} = 0 \implies f(y) = \text{constant } C. \qquad f \textit{ depends on } y \textit{ alone}$$

Hence

$$U = t^2 y + t^3 + C.$$

The ODE is

$$\mathrm{d}U = 0$$

and integrating yields

$$t^2 y + t^3 + C = C_1,$$

so

$$t^2 y + t^3 = C_2 \tag{4.9}$$

is the solution of (4.3). Eq. (4.9) identically satisfies (4.3) and is free of derivatives.

1.1 The same U from the other integration order

From (4.7),

$$\frac{\partial U}{\partial y} = t^2 \implies U = t^2 y + g(t). \qquad \textit{integrate with respect to } y \textit{ this time}$$

Here $g(t)$ is constant of integration and independent of y, but may depend on t. Substituting U into (4.6) yields

$$\frac{\partial}{\partial t}(t^2 y + g(t)) = 2ty + \frac{\mathrm{d}}{\mathrm{d}t} g(t) = 2ty + 3t^2 \qquad \textit{compare with (4.6)}$$

$$\frac{\mathrm{d}g(t)}{\mathrm{d}t} = 3t^2$$

$$g(t) = t^3 + C. \qquad \textit{integrate}$$

Hence

$$U = t^2 y + t^3 + C$$

etc. So, we could have reversed the order of using (4.6) and (4.7) and still have obtained the same U.

2 Exercises: test for exactness, and if so, solve

Put as: $M\ \mathrm{d}t + N\ \mathrm{d}y = 0$.

Source: Spiegel, *Applied Differential Equations*, p. 32.

Example 4.2 ((a) $3t\ \mathrm{d}t + 4y\ \mathrm{d}y = 0$).

$$3t\ \mathrm{d}t + 4y\ \mathrm{d}y = 0 \tag{4.10}$$

$$M = 3t, \qquad N = 4y.$$

Here $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t} = 0$ so the ODE (4.10) is exact.

$$\frac{\partial U}{\partial t} = 3t \implies U = \int 3t\ \mathrm{d}t = \frac{3}{2}t^2 + f(y) \qquad \textit{integrate with respect to } t$$

$$\frac{\partial U}{\partial y} = \frac{\mathrm{d}f(y)}{\mathrm{d}y} = 4y \implies f(y) = 2y^2 + C_1 \qquad \textit{match } N\textit{; integrate}$$

$$U = \frac{3}{2}t^2 + 2y^2 + C_2$$

ODE is $\mathrm{d}U = 0$ + integrating yields

$$\frac{3}{2}t^2 + 2y^2 + C_2 = C,$$

so

$$3t^2 + 4y^2 = C$$

is solution.

Example 4.3 ((b) $y' = \dfrac{t-y}{t+y}$).

$$y' = \frac{t-y}{t+y} \implies \frac{\mathrm{d}y}{\mathrm{d}t} = \frac{t-y}{t+y} \implies (t-y)\ \mathrm{d}t - (t+y)\ \mathrm{d}y = 0 \qquad \textit{cross-multiply and collect}$$

$$\frac{\partial U}{\partial t} = M = t - y, \qquad \frac{\partial U}{\partial y} = N = -(t+y)$$

$$\frac{\partial M}{\partial y} = -1, \qquad \frac{\partial N}{\partial t} = -1 \qquad \textit{so ODE is exact}$$

$$\frac{\partial U}{\partial t} = t - y \implies U = \frac{t^2}{2} - yt + f(y) \qquad \textit{integrate with respect to } t$$

$$\frac{\partial U}{\partial y} = \frac{\partial}{\partial y}\Big(\frac{t^2}{2} - yt + f(y)\Big) = -t + \frac{\mathrm{d}f(y)}{\mathrm{d}y} = -t - y \qquad \textit{match } N$$

$$\implies \frac{\mathrm{d}f(y)}{\mathrm{d}y} = -y \implies f(y) = -\frac{y^2}{2} + C$$

$$U = \frac{t^2}{2} - yt - \frac{y^2}{2} + C$$

$$t^2 - 2yt - y^2 = C \quad \text{is solution.}$$

Example 4.4 ((c) $2ty\,y' = t^2 - y^2$).

$$2ty\,y' = t^2 - y^2$$
$$2ty\,\frac{\mathrm{d}y}{\mathrm{d}t} = t^2 - y^2$$

$$(t^2 - y^2)\,\mathrm{d}t - 2ty\,\mathrm{d}y = 0 \tag{4.11}$$

$$M = \frac{\partial U}{\partial t} = t^2 - y^2, \qquad N = \frac{\partial U}{\partial y} = -2ty$$
$$\frac{\partial M}{\partial y} = -2y, \qquad \frac{\partial N}{\partial t} = -2y$$
$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t} \implies \text{ODE (4.11) is exact}$$
$$\frac{\partial U}{\partial t} = t^2 - y^2 \implies U = \int (t^2 - y^2)\,\mathrm{d}t = \frac{t^3}{3} - y^2 t + f(y)$$
$$\frac{\partial U}{\partial y} = \frac{\partial}{\partial y}\Big(\frac{t^3}{3} - y^2 t + f(y)\Big) = -2yt + \frac{\mathrm{d}f(y)}{\mathrm{d}y} = -2ty \qquad \textit{match } N$$
$$\frac{\mathrm{d}f(y)}{\mathrm{d}y} = 0 \implies f(y) = C$$
$$U = \frac{t^3}{3} - y^2 t + C$$
$$\therefore \quad t^3 - 3y^2 t = C \quad \text{is solution.}$$

Example 4.5 ((d) $y' = \dfrac{t}{t+y}$ — not exact).

$$y' = \frac{t}{t+y} \implies t\,\mathrm{d}t - (t+y)\,\mathrm{d}y = 0$$
$$M = \frac{\partial U}{\partial t} = t, \qquad N = \frac{\partial U}{\partial y} = -(t+y)$$
$$\frac{\partial M}{\partial y} = 0 \quad \text{(i)}, \qquad \frac{\partial N}{\partial t} = -1 \quad \text{(ii)}$$
$$\therefore \quad \textbf{Not exact.}$$

3 Making an inexact ODE exact by multiplying with an integrating factor

Example 4.6 ((e) $y\,\mathrm{d}t + (3 + 3t - y)\,\mathrm{d}y = 0$).

$$y\,\mathrm{d}t + (3 + 3t - y)\,\mathrm{d}y = 0 \tag{4.12}$$

$$M = y, \qquad N = 3 + 3t - y$$
$$\frac{\partial M}{\partial y} = 1, \qquad \frac{\partial N}{\partial t} = 3$$
$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial t}$$

so (4.12) is **NOT** exact. Note that

$$\frac{1}{M}\Big(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}\Big) = \frac{1}{y}\,(3-1) = \frac{2}{y} = g(y)$$

is a function of y only and not also t. Then

$$\mathrm{e}^{\int g(y)\,\mathrm{d}y} = \mathrm{e}^{\int \frac{2}{y}\,\mathrm{d}y} = \mathrm{e}^{2\ln|y|} = y^2$$

is an integrating factor.

Multiply eq. (4.12) by y^2:

$$y^3\,\mathrm{d}t + y^2\,(3 + 3t - y)\,\mathrm{d}y = 0 \tag{4.13}$$

$$\begin{aligned} M &= y^3, & N &= 3y^2 + 3t\,y^2 - y^3 \\ \frac{\partial M}{\partial y} &= 3y^2, & \frac{\partial N}{\partial t} &= 3y^2 \end{aligned}$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$, the new differential eq. (4.13) is exact.

Theorem 4.2 (Integrating factor, Theorem 1). If

$$\frac{1}{N}\Big(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial t}\Big) = f(t),$$

a function of t only, then $\mathrm{e}^{\int f(t)\,\mathrm{d}t}$ is an integrating factor.

Theorem 4.3 (Integrating factor, Theorem 2). If

$$\frac{1}{M}\Big(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}\Big) = g(y),$$

a function of y only, then $\mathrm{e}^{\int g(y)\,\mathrm{d}y}$ is an integrating factor.

Source: Murray Spiegel, *Applied Differential Equations.*

Note. Equation (4.13) is the equation set as Tutorial 4: "Regarding last page of posted notes *Making inexact ODE exact by multiplying with an integrating factor*, find the solution of Eq. (13)." The professor's own solution continues the numbering of this run with (13)–(15) and is typed in the Tutorial 4 solutions.